Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/24/2020
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rweber@matsdev.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
PI Address 1:	3090 N Daniels Ct		
PI Address 2:			
PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed project will fund the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) project [1] to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Under the proposed project, NASA investigators will implement experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. Flight experiments on molten barium oxide-titanium dioxide pseudo-binary compositions will be performed during the first and second years of the project. Samples will be launched on H-II Transfer Vehicle-8 (HTV-8) in mid-2019 and a second set of samples is planned for launch on HTV-9 in 2020. The measured thermophysical properties (density, viscosity, and surface tension) will be correlated with atomic structure obtained from synchrotron x-ray and neutron methods and by using structure-property models in collaboration with the Japanese investigators. The third year will be devoted to analysis of flight data, completing characterization of samples, and publishing the results. Work will be published in a timely way consistent with the goal of providing data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property data that can be used to optimize models used to design materials from first principles.
Task Progress:	This project is part of a collaborative international JAXA project to investigate the: (i) properties of liquid phase and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) [1], and (ii) atomic structure using ground-based synchrotron and neutron studies. The NASA investigators are supplying samples for experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. In addition to the scientific research, the proposed project will help to establish details of working with JAXA to provide flight samples and help to characterize the ELF capabilities by measuring US investigator materials. Samples were launched to ISS on SPX-20 in spring of 2020. Currently, ground based work has been focused on using DL-Poly to model structure and dynamics of the liquids. This approach will be used to compare measured values with the model in order to improve predictive capabilities for liquid properties and structure.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Alderman OLG, Benmore CJ, Neuefeind J, Tamalonis A, Weber R. "Molten barium titanate: a high-pressure liquid silicate analogue." J Phys Condens Matter. 2019 May 22;31(20):20LT01. https://doi.org/10.1088/1361-648X/ab0939 ; PMID: 30790768 , May-2019
Articles in Peer-reviewed Journals	Alderman OLG, Benmore CJ, Tamalonis A, Weber R. "Rare-earth titanate melt structure and glass formation." Int J Appl Glass Sci Technol. 2019 Oct;10(4):463-78. https://doi.org/10.1111/ijag.13479 , Oct-2019